

LeatherBoard: Sustainable On-body Rapid Prototyping with Leather Scraps and Machine Embroidery

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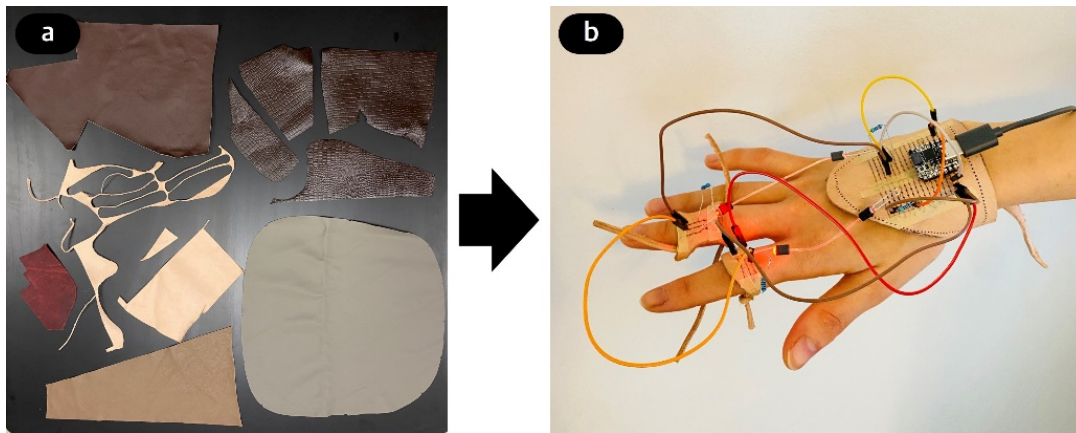


Figure 1: LeatherBoard. (a) Discarded leather scraps, (b) finger strain sensor prototyped on the LeatherBoard

ABSTRACT

The increasing demand for wearable electronics calls for more effective prototyping methods that deliver reliability while maintaining user comfort and versatility. Traditional methods often rely on rigid materials, leading to skin discomfort and challenges in rapid prototyping. Meanwhile, the fashion and electronics industries contribute to environmental issues by generating significant waste, including tons of leather scraps every year. This paper explores the potential of these leather scraps as a resource for creating flexible breadboards. We introduce LeatherBoard, a sustainable approach that uses machine embroidery with conductive threads for efficient and standardized prototyping. Our contributions include the development of LeatherBoard and its evaluation in terms of electrical reliability and durability. We also demonstrate its potential through three prototyped applications, showcasing LeatherBoard's strength as a prototyping platform.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI);

KEYWORDS

Rapid prototyping, On-body interface, Sustainability, Leather

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1 INTRODUCTION

As the market for wearable electronics grows, there is a clear need for improved prototyping methods to efficiently produce reliable prototypes. Essential qualities like flexibility, comfort, and versatility are often compromised in many current methods, which tend to use rigid materials that can cause skin discomfort or fail to align with standard rapid prototyping resources. Furthermore, the thickness and weight of these materials often make them unsuitable for integration with textiles or direct application on the body, where wearable electronics are increasingly accommodated. This highlights the necessity for innovative solutions in wearable electronics prototyping.

Meanwhile, the fashion and electronics industries are significant contributors to environmental waste, a concern aggravated by the disposal of leather scraps from product manufacturing. Recognizing leather's inherent potential as a flexible skin material, this paper employs a research-through-design approach [8] to investigate repurposing leather scraps for wearable electronics prototyping. We introduce LeatherBoard, developed using machine embroidery with conductive threads. This innovative platform not only aids in effective wearable device development but also addresses the

pressing issue of leather and electronic waste (Figure 1). The paper details LeatherBoard’s unique contributions in this emerging field as follows:

- The LeatherBoard represents a sustainable method for prototyping on-body electronics, utilizing discarded leather scraps for a more efficient prototyping platform.
- This approach incorporates machine embroidery with conductive threads, enabling quick and semi-automated fabrication in line with industry standards.
- The use of leather scraps offers convenience in rapid prototyping due to their ready availability on human body.
- The paper evaluates the reliability of this leather scrap-based breadboard in terms of electrical connectivity and durability.
- We showcase the potential of the prototyping platform through three applications, demonstrating LeatherBoard’s effectiveness and versatility.

2 RELATED WORK

2.1 Rapid prototyping and machine embroidery for on-body electronics

The form factor of on-body electronics is crucial for their performance, comfort, and durability, as the human body surface is full of curvatures and deformable. It also has been taken into account during the prototyping phase to establish a reliable basis of the development process [7]. The ideal prototyping platform for wearable electronics may achieve conformability [3, 7, 13, 18], light weight [19, 20, 25], modularity [5, 17, 22], compatibility with industry standards for electronic parts [21, 28, 29, 34], and affordability [21, 33]. Breadboard Bracelet [35] is a straightforward way to convert standard breadboards into a wearable form factor, but the rigidity of the plastic board limits its conformability to the body surface. FlexBoard [21] demonstrated a successful transformation of traditional breadboard for conformability to curvatures. However, the 3D printed structure using a PLA filament and metal strips can introduce considerable weight and discomfort when directly attached to the skin. Prototyping platforms that use flexible materials, such as Wearable Bits [17] or Breadboard Pincushion [26] can offer better comfort, but they showed limitations in mass fabrication and compatibility with rapid prototyping components [3]. JellyBoard [28] allowed the use of off-the-shelf rapid prototyping components by implementing female pin headers in silicone, but has production and durability issues related to the copper strips. LeatherBoard, on the other hand, displays both the flexibility of textiles and standardized pin spacings from machine embroidery.

Meanwhile, as e-textiles continue to rise in popularity, embroidery has also been actively explored due to its vast fabrication techniques and long history, which can be beneficial in creating on-body interfaces [16]. Computerized electronic embroidery has been found to be advantageous for both aesthetics and standardization in production across various areas in wearable electronics, from sensors and actuators to power generators [10, 15, 24, 32]. TexYZ [1] demonstrated mutual capacitive sensors for textiles using machine-embroidered enamel wires as touch electrodes. Similarly, SkinLace [14] fabricated a conductive lace by machine embroidery for on-skin circuitry for displays, electrodes, and antennas. In Embr [6], a conductive threads was hand-embroidered to function as a

resistive heater, which then activated the thermochromic liquid crystal display through the heat uniquely generated by each embroidery pattern. FabriClick [9] employed machine embroidery to create circuit and touch electrodes, as well as a three-dimensional structure on a stretchy textile to enhance usability. While machine embroidery has been actively adopted to create flexible textile-based circuits, the holes created by the needles have been interpreted as something to be minimized so as not to damage the textile, but not yet explored as a potential for accommodating electronics.

2.2 Creating electrical connections in rapid prototyping

The ability to assist to quickly create electrical connections is the fundamental requirement for prototyping platforms like breadboards. While soldering is a highly reliable and widely utilized method to create electrical connections, it poses limitations in re-configurability of systems [33]. This flexibility is essential in the development of wearable systems, where adjusting the positioning of components for optimal layout on the body surface is a critical consideration. Traditional breadboards utilize metal strips as in FlexBoard [21] and JellyBoard [28]. In CurveBoard [34] and Rubber Breadboard [2], conductive silicone replaced the metal strips, but it also showed deformations that resulted in changes in resistance over time. ThreadBoard [13] and SkinKit [22] used magnetic connections featuring convenience, but they were not compatible with electronic parts commonly used for rapid prototyping. Sketch & Stitch [11] and Wearable Bits [17] have demonstrated textile-friendly options such as conductive threads and snap buttons, yet their electrical connection can easily become loose. Printing with conductive materials provides design freedom and fabrication convenience, while creating electrical connections with other parts requires additional methods, such as conductive adhesives or conductive thread [4, 20]. The LeatherBoard system described in this study features pinholes similar to those found in traditional breadboards, with conductive threads constructing electrical connections.

2.3 Sustainability in Fashion and Electronics

The fashion and textile industry are widely recognized as a significant contributor to the global environmental crisis. Along with water pollution and greenhouse gas emissions, low recycling rates and discarded textile scraps during manufacturing pose threats to the earth. The leather industry holds a significant position in the world, and the annual amount of global scrap leather waste is about 2 million tons, most of which is incinerated or landfilled [36]. Meanwhile, global electronic waste (e-waste) is the fastest-growing among solid wastes, having reached 53.6 million tonnes in 2019 [31]. Prototyping electronics may play a role in the growth of e-waste, and it is time to consider its environmental impacts. Sustainability is gaining attention within the field of human-computer interaction (HCI), with most studies focused on responsible production and consumption [12]. While there are interests in hunger, infrastructure, and clean energy, it is important to address the issue of e-waste. Projects such as the production of biodegradable materials from bacterial cellulose [30], the development of a modular toolkit for sustainable package design for prototyping [23], and the utilization

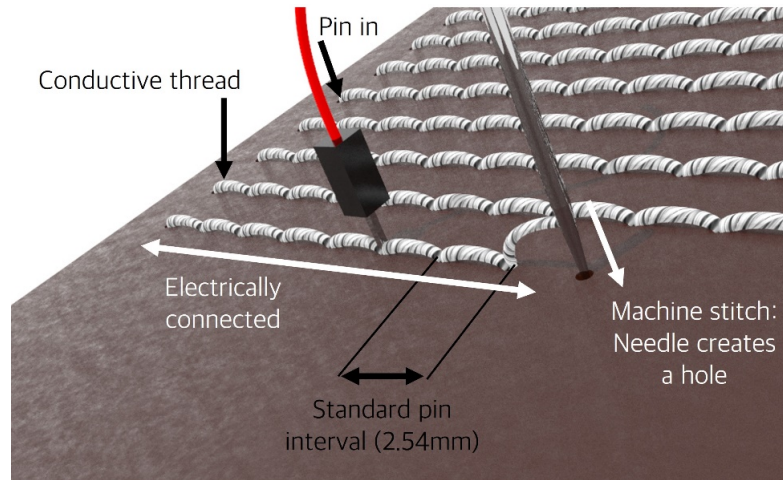


Figure 2: Design of the LeatherBoard

of spent coffee grounds for 3D printing [27] have contributed to the expansion of the field of sustainable prototyping.

3 LEATHERBOARD

3.1 Design

LeatherBoard aims to use discarded leather scraps from commercial leather product manufacturers to create a flexible, durable, and sustainable breadboard for prototyping on-body electronics. We utilized leather scraps that are too small or have surface defects, such as scratches, which are not suitable for commercialization. Machine embroidery created holes and stitches on the leather at the same intervals as the pin spacing (2.54 mm) of traditional breadboards. When a needle penetrates leather, it creates a hole that allows a pin or electrode of an electronic component to be inserted. In contrast to other textiles made of yarns or fibers where the hole may loosen the structure, in leather, the hole provides a tight fit to securely hold the pin. A conductive thread (Madeira HC40 [37], $<300 \Omega/\text{m}$) is used on both sides (upper and bobbin) to establish electrical connections between the holes. This thread serves the same purpose as the metal strip in traditional breadboards. When a pin is inserted into a hole, the pin and the conductive thread are mechanically clamped within the hole, creating a stable electrical connection (Figure 2). The LeatherBoard allows users to select an appropriate leather scrap based on the target body surface and design the necessary holes and stitches. Unlike traditional breadboards, which are limited to only five holes connected to each other due to the length limit of metal strips, the LeatherBoard can extend the length of the electrical connection as long as the leather scrap allows. This feature is particularly useful when the application needs to cover a large area of the body surface like torso. LeatherBoard consists of two layers of leather: one with conductive stitch lines and the other with nothing but solely for protection purposes. The leather is strong enough to block sharp pins or electrodes and protect the skin from potential pain.

3.2 Fabrication

Fabrication consists of the following steps (Figure 3):

- Measure the available area based on the shape of the leather scrap and the target body part.
- Design the embroidery shape. A rectangle is recommended, but any other shape is acceptable as long as it allows parallel rows. Ensure that the stitch length corresponds to the standard pin spacing of traditional breadboards. Machine embroidery involves a continuous line stitch, which requires connections between rows unavoidably.
- Load a layer of water-soluble stabilizer. Since leather is often too thick to be clamped in the embroidery frame, the stabilizer serves as a base for the leather.
- Place the leather on the stabilizer and begin the embroidery process. It is important to manually hold the leather in place to prevent it from shifting on the stabilizer. After a few stitches, it will remain in place.
- Once the embroidery is complete, remove the leather and dissolve the stabilizer in water.
- Cut the thread connections between the rows to ensure each row works independently.
- Tape the edges to prevent the cut edges of the threads from touching the other row. Stitch another layer of leather on the bottom to protect the skin from pins.
- Plug the electronic components into the holes.

4 RELIABILITY EVALUATION

4.1 Electrical connection

In LeatherBoard, the electrical connection relies on the mechanical contact between the pin and the conductive thread in a hole. To determine if plugging a pin into any hole guarantees electrical contact, we tested all the holes within a sample board containing approximately 100 holes. A LeatherBoard with 9 holes * 12 rows and two male-male jumper wires were utilized to test electrical connectivity. One jumper wire was plugged into the end hole of a row, while the other was plugged into all other holes of the same

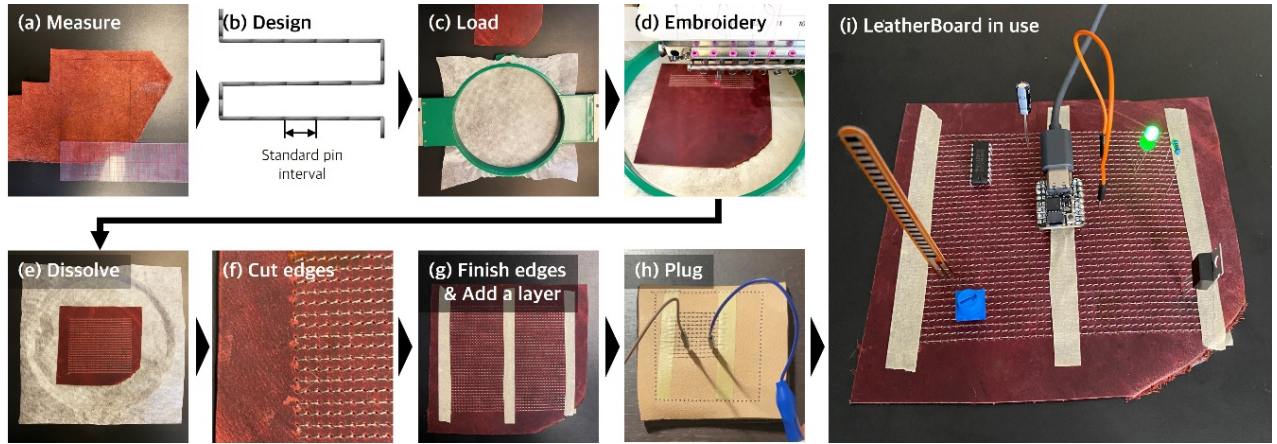


Figure 3: Fabrication process of LeatherBoard

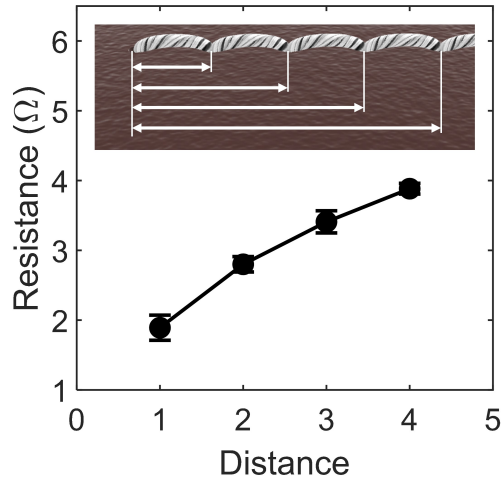


Figure 4: Resistance by pinhole distance

row. Using a multimeter that beeps when the resistance is lower than $30\ \Omega$, we checked the connection between the other ends of the jumper wires. We identified that all holes (8 holes * 12 rows) were electrically connected to the end hole of the row.

4.2 Resistance between pins

A LeatherBoard with 5 holes * 12 rows and two male-male jumper wires were utilized. Following the methodology outlined in section 4.1, one jumper wire was inserted into the end hole of a row, while the other was inserted into all other holes of the same row. We then proceeded to measure the resistance between the ends of the jumper wires. The mean and standard deviation of resistance were $3.88 \pm 0.27\ \Omega$, $3.41 \pm 0.55\ \Omega$, $2.80 \pm 0.38\ \Omega$, and $1.89 \pm 0.62\ \Omega$ for distances ranging from four holes apart (end to end) to adjacent holes (next to each other) as shown in Figure 4. These values were considerably lower than those of CurveBoard [34].

4.3 Durability

Twelve pairs of holes and two male-male jumper wires were used to test the conductive thread's durability. The jumper wire was plugged into a hole of a pair and then repeatedly plugged and unplugged until the digital multimeter indicated disconnection ($>30\ \Omega$). On average, the conductive thread disconnected or broke after 135.42 ± 51.46 plug & unplug cycles. The minimum number of cycles was 46 and the maximum was 232. The median was 138.5.

5 APPLICATIONS

5.1 Finger strain sensor using stretchable fiber optics

We discovered that many leather scraps have a net-like appearance after the panel for commercial products had been cut out (Figure 1a, scraps in bright beige). These scraps are identified as being used as strings to hold the leather around the body, such as on the hands and fingers (Figure 5a). We prototyped a finger strain sensor to potentially track the movement of two fingers by measuring the changes in light intensity through a pair of stretchable fiber optics. This prototype required a microcontroller (Adafruit QT Py), two LEDs, and two photosensors. The straps facilitated fast and convenient prototyping as we did not have to search for a way to attach the leather boards to the body. The main sensing body was located on the large panel on the back of the hand, while the small panels on the fingers accommodated LEDs. This allowed for easier determination of whether stretchable fiber optic works for strain sensing and the optimal length for optimal sensing performance.

5.2 Arm capacitive touch panel

A LeatherBoard was used to cover the lower arm and serve as a platform for a capacitive touch sensing panel (Figure 5b). Three buttons made of conductive embroidery were placed on the edge of the LeatherBoard where there are no holes. One of the common issues in many conductive thread-based prototyping is the connection between the thread and hard electronic parts, which can easily become loose. However, in this case, a conductive thread from the capacitive touch buttons created knots on the stitches on

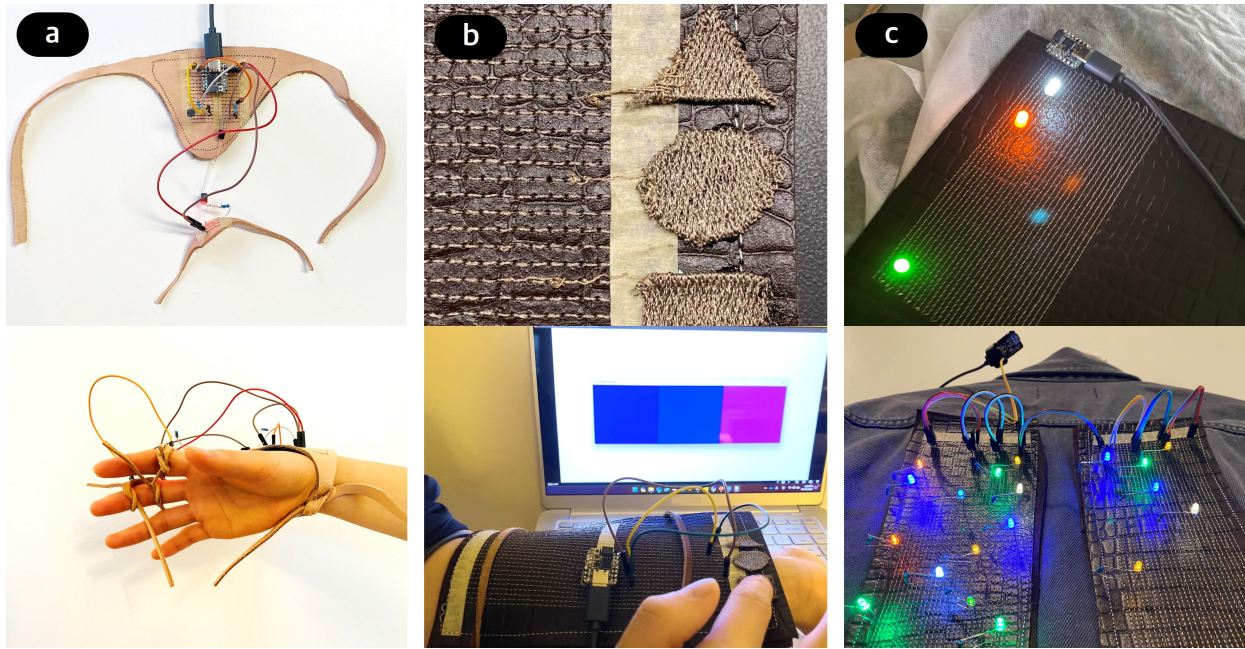


Figure 5: Applications prototyped on the LeatherBoard

the LeatherBoard, ensuring secure connections ensuring secure connections as thread-thread knots were formed. If the back side of the capacitive touch buttons is well insulated, it is possible to rearrange the buttons even on the LeatherBoard to determine the optimal layout on the lower arm for an improved user experience.

5.3 Collective design of aesthetic costume using LEDs

Designing aesthetic costumes with electronics can be challenging due to the difficulty of testing layouts on the garment: traditional breadboards are often too heavy and small to explore various design possibilities on a large body surface, especially when designing collaboratively. To address this issue, we utilized two large panels that cover a significant portion of the back of a jacket and arranged the anode and cathode from the power source. The LEDs in various colors were placed according to each designer's preference and adjusted until a satisfactory outcome was achieved (Figure 5c).

6 DISCUSSION

6.1 Challenges

Removing the thread connection between rows is still not desirable due to the labor-intensive process and the risk of the cut edge of stitches touching the adjacent row and causing a short circuit. While it is possible to program a cut command at the end of each row, controlling the cut edge of the thread remains a challenge. Another challenge related to the thread is its abrasion durability. The reliability test showed that the thread can withstand over 130 cycles of pin plugging and unplugging in general. However, the minimum number of cycles was less than 50, which could be critical. To protect the conductive thread from strong frictions, it is possible

to use an abrasion-resistant thread such as nylon on the upper side and use the conductive thread only on the bobbin side. This approach may degrade the connectivity between the pin and the thread. To enhance durability against abrasions, it will be necessary to conduct a careful investigation and explore other materials and tension settings.

Additionally, the thickness of the LeatherBoard may become a concern when the prototype is involved with active body movements. While the thin form factor is advantageous, most pins in electronic components are designed for traditional breadboards, leaving most of the pin outside of the LeatherBoard. Although structured evaluation was not adopted for the LeatherBoard's holding force of electronic components, it was observed that a hole with a pin of a jumper wire can hold a small metal wrench (Figure 6). We also observed that the board can secure the plugged pins when moderately flexed. However, additional structured evaluations using a force meter under various conditions is necessary to determine the feasibility of the electrical connection in real on-body situations. An additional sponge layer sandwiched between the two leather layers helped to hold the pins more securely by allowing them to be plugged deeper. However, this study did not adopt the sponge method due to the increased thickness, as the two leather layers are already considerable in terms of skin comfort.

6.2 Opportunities

Although small panels were used in this study, LeatherBoard can be easily connected to other panels by sewing to cover a larger surface area. This means it can be used on curved parts of the body, such as knees or elbows, if the boards with multiple panels construct the curvy shape. The current version with two leather layers is not very stretchable, but it can be made stretchable by

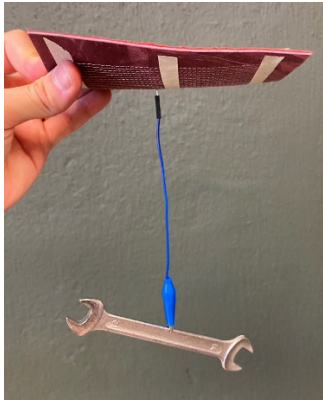


Figure 6: LeatherBoard holding a small metal wrench

giving slits between the rows. The LeatherBoard's thin form factor makes it suitable for various applications beyond wearable technology, including narrow spaces such as pipes. Furthermore, the fabrication technique used for the LeatherBoard can be applied to create interactive leather accessories, such as wallets or bags, by selecting compatible electronic components.

This study sourced leather scraps that were either small leftovers after cutting or relatively large pieces with defects such as scratches that are not allowable for commercial products. While large pieces may be useful for covering larger areas of the body, such as the torso, it would be more meaningful to use smaller pieces that have no further potential for commercialization, as smaller pieces may serve functionality as strings (as shown in Figure 5a), conform to body curvatures, or provide breathability. To make this method more accessible, computer vision techniques can be used to identify available areas and create stitch designs when using informal shapes. The presented method in this study includes consumable components, such as water-soluble stabilizer, which can be collected and reused in a mass production setting or replaced by another component, such as the sponge layer discussed above.

7 CONCLUSION

This paper presents LeatherBoard, a sustainable method for prototyping on-body applications using machine embroidery. We utilized discarded leather scraps, conductive threads, and an embroidery machine to create stitches on the leather at a standardized interval, allowing for easy installation of electronic components available in the market. The flexible form factor facilitates easy prototyping on the body, while the net-like structure of scraps provides straps for securing the device. Three applications were prototyped on LeatherBoard and demonstrated the effectiveness and versatility: a finger strain sensor, an arm capacitive touch panel, and a collectively designed aesthetic costume using LED.

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